AP PHYSICS 1

Unit 1 • Self Check 1.1 • Topics 1.1 to 1.3

Self Check 1.1 • Topics 1.1 to 1.3

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

Which pair contains only vector quantities?

(A) Displacement and velocity
(B) Speed and velocity
(C) Distance and acceleration
(D) Time and force

Question 2

A cyclist rides 6 km east, then 2 km west, in 0.5 h. What is the average velocity?

(A) 16 km/h east
(B) 8 km/h east
(C) 8 km/h west
(D) 4 km/h east

Question 3

A car moving in the negative x-direction is slowing down. Which signs are correct?

(A) $v < 0, a < 0$
(B) $v > 0, a < 0$
(C) $v < 0, a > 0$
(D) $v > 0, a > 0$

Question 4

An object starts from rest and accelerates at $4\,\text{m/s}^2$ over a distance of 18 m. What is its final speed?

(A) 6 m/s
(B) 72 m/s
(C) 144 m/s
(D) 12 m/s

Question 5

On a velocity-time graph, the area between the graph and the time axis represents the

(A) acceleration
(B) displacement
(C) speed
(D) time

Question 6

The position-time graph of an object is a horizontal line. The object is

(A) at rest
(B) moving with constant acceleration
(C) moving at a constant nonzero velocity
(D) slowing down

Question 7 (Free response)

An elevator starts from rest and moves up. It reaches 3 m/s in 2 s, moves at 3 m/s for 4 s, then stops in 3 s.

(a) Find the acceleration during each part.
(b) Find the total displacement.
(c) Find the average velocity for the whole trip.